Single Image Dehazing using a Model-Agnostic Convolutional Neural Network

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***Abstract*—Image dehazing is an important preprocessing step in computer vision applications affected by environmental conditions such as fog and haze. Traditional approaches rely heavily on physical models of atmospheric scattering, which often require estimation of intermediate parameters and may not generalize well to real-world scenarios. In this project, we implement a simplified version of a Generic Model-Agnostic Network (GMAN), a deep learning-based approach that directly learns a mapping from hazy images to clean images without relying on explicit physical models. The model is trained end-to-end using synthetic datasets and evaluated using standard image quality metrics.**

I. Introduction

Many real-world vision systems, including surveillance, autonomous driving, and remote sensing, are significantly affected by atmospheric conditions such as haze and fog. These conditions degrade image quality by reducing contrast and visibility, making downstream tasks such as detection and recognition more challenging.

Single image dehazing aims to recover a clear image from a single hazy input. Traditional approaches rely on the atmospheric scattering model, which describes the formation of haze in images. However, these methods require estimation of unknown parameters such as transmission maps and atmospheric light, making the problem ill-posed.

With the advancement of deep learning, data-driven approaches have emerged as a powerful alternative. Instead of explicitly modeling the physical process, neural networks can learn a direct mapping between hazy and clean images. In this project, we explore such an approach by implementing a model-agnostic convolutional neural network for image dehazing.

II. Related Work

Image dehazing methods can broadly be categorized into traditional and learning-based approaches.

Traditional methods rely on prior assumptions about image properties. One of the most influential works is the Dark Channel Prior (DCP), which assumes that in most non-sky patches, at least one color channel has very low intensity. Other methods use color attenuation priors or non-local priors to estimate haze-related parameters.

Learning-based approaches leverage machine learning and deep neural networks. Early methods used shallow learning techniques, while recent works employ convolutional neural networks (CNNs) to estimate transmission maps or directly restore clean images. Models such as DehazeNet, MSCNN, and AOD-Net follow the plug-in approach by estimating intermediate parameters.

More recent approaches adopt end-to-end learning strategies that bypass explicit modeling of atmospheric parameters. The Generic Model-Agnostic Network (GMAN) falls into this category, directly learning the transformation from hazy to clear images.

III. Methodology

A. Problem Formulation

The traditional atmospheric scattering model is given by:

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{1}$$

where $I(x)$ is the observed hazy image, $J(x)$ is the clean image, $t(x)$ is the transmission map, and A is the global atmospheric light.

Instead of estimating $t(x)$ and A, the proposed approach directly learns a mapping:

$$J(x) = f(I(x)) \tag{2}$$

where $f(\cdot)$ is a convolutional neural network.

B. Network Architecture

The proposed model follows an encoder-decoder architecture with residual learning. The overall architecture of the proposed model is illustrated in Fig. 1. The key components include:

- **Encoder:** Extracts features from the input hazy image.
- **Residual Blocks:** Improve feature learning and convergence.
- **Decoder:** Reconstructs the dehazed image.
- **Global Residual Connection:** Preserves image details.

As shown in Fig. 1, the model leverages both local and global residual learning to effectively remove haze while preserving image details.

C. Detailed Network Design

The network consists of multiple convolutional layers organized in an encoder-decoder framework. The encoder progressively reduces the spatial resolution while increasing feature depth, allowing the network to capture high-level contextual information. Specifically, convolutional layers with a stride of 2 are used for downsampling.

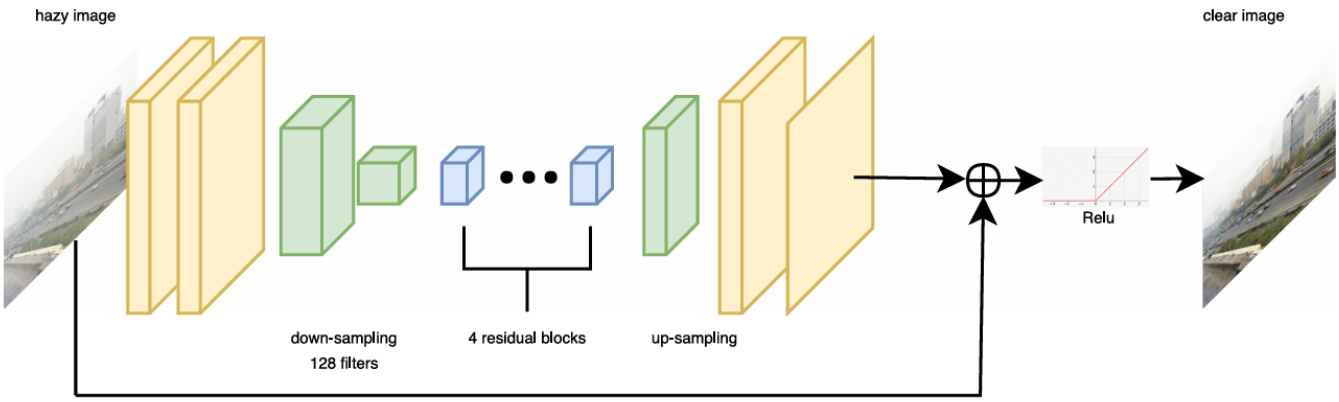

Fig. 1. Architecture of the proposed Generic Model-Agnostic Network (GMAN) for single image dehazing. The network follows an encoder-decoder structure with down-sampling and up-sampling layers. Multiple residual blocks are used in the intermediate stage to enhance feature learning. A global residual connection is applied by adding the input hazy image to the network output, followed by a ReLU activation to produce the final dehazed image.

The intermediate representation is processed using a series of residual blocks. Each residual block consists of two convolutional layers with ReLU activation and a skip connection. This design helps mitigate the vanishing gradient problem and enables the network to learn deeper representations effectively.

The decoder mirrors the encoder structure by using transposed convolutional layers for upsampling. This allows reconstruction of the spatial resolution while preserving learned features. The final output is combined with the input image using a global residual connection, ensuring that the model focuses on learning the haze component rather than reconstructing the entire image from scratch.

D. Residual Learning

Residual learning is applied both locally and globally. Local residual blocks help in learning feature representations effectively, while the global residual connection ensures that the network focuses on learning the difference between hazy and clean images.

E. Loss Function

The model is trained using Mean Squared Error (MSE) loss:

$$L = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{J}(x_i) - J(x_i) \right\|^2 \tag{3}$$

where $\hat{J}(x_i)$ is the predicted output and $J(x_i)$ is the ground truth.

F. Dataset

The model is trained using the RESIDE dataset, specifically the Indoor Training Set (ITS). The Synthetic Objective Testing Set (SOTS) is used for evaluation. Due to computational constraints, a subset of the dataset is used in this implementation.

G. Training Details

The model is trained using the Adam optimizer with an initial learning rate of 0.001. A batch size of 4 is used to balance computational efficiency and memory constraints. The training is conducted for a limited number of epochs due to hardware limitations. All input images are resized to a fixed resolution of 224×224. The loss function used is Mean Squared Error (MSE), which measures the pixel-wise difference between the predicted and ground truth images. To improve training efficiency, a subset of the dataset is used. This subset still captures sufficient variability in haze conditions while significantly reducing training time.

H. Evaluation Metrics

To quantitatively evaluate the performance of the model, two commonly used metrics are employed: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM).
PSNR measures the reconstruction quality of the dehazed image compared to the ground truth, with higher values indicating better performance. SSIM evaluates perceptual similarity by considering structural information, luminance, and contrast.
These metrics provide a comprehensive assessment of both numerical accuracy and visual quality.

IV. Results and Discussion

The proposed model was evaluated on the SOTS dataset using two standard metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). The obtained results are:

- **PSNR:** 17.94 dB
- **SSIM:** 0.7133

These results are compared with existing state-of-the-art methods as reported in the literature. Table I presents a comparison of performance across different approaches.

TABLE I
PERFORMANCE COMPARISON ON SOTS DATASET

Method	PSNR	SSIM
DCP	18.87	0.794
DehazeNet	22.66	0.833
MSCNN	20.01	0.791
AOD-Net	21.01	0.837
GFN	22.44	0.884
GMAN (Original)	27.94	0.897
Proposed (Ours)	**17.94**	**0.7133**

A. Analysis of Results

From Table I, it can be observed that the performance of the implemented model is lower than that of the original GMAN and other deep learning-based methods. While the model demonstrates the capability to reduce haze and improve visual quality, there is a noticeable gap in quantitative performance.

The primary reasons for this performance difference are as follows:

1) Limited Training Data: The original GMAN model is trained on the full RESIDE dataset, which contains a very large number of hazy images generated under diverse conditions. In contrast, due to computational and storage constraints, this implementation uses only a small subset of the dataset (approximately 1000 images). This significantly limits the model's ability to generalize across varying haze densities and scene structures.

2) Reduced Training Duration: Deep learning models typically require extensive training over many epochs to achieve optimal performance. In this project, the model was trained for a limited number of epochs, which restricts convergence and prevents the network from fully learning the underlying mapping between hazy and clean images.

3) Simplified Network Architecture: The implemented architecture is a simplified version of the original GMAN model. The original network contains a larger number of residual blocks and more complex feature extraction layers, enabling better representation learning. The reduced complexity in this implementation leads to lower expressive power and consequently lower performance.

4) Resolution Constraints: All images were resized to 224×224 to accommodate computational limitations. This resizing leads to loss of fine details, which negatively impacts reconstruction quality and reduces both PSNR and SSIM values.

5) Lack of Advanced Training Strategies: State-of-the-art models often incorporate additional techniques such as learning rate scheduling, data augmentation, perceptual loss functions, and multi-scale training. These enhancements were not included in this implementation, further contributing to the performance gap.

B. Qualitative Observations

Despite the lower quantitative performance, the model is able to visibly improve image clarity and reduce haze in most cases. The output images exhibit enhanced contrast and improved visibility compared to the input hazy images. However, some residual haze and loss of fine details can still be observed, particularly in regions with dense haze. As shown in Fig. 2, the proposed model is able to reduce haze and enhance visibility in the scene. However, some residual haze and slight loss of details can still be observed compared to the ground truth.

Fig. 2. Qualitative comparison of dehazing results. From top to bottom: input hazy image, output of the proposed model, and ground truth clean image. The model improves visibility and contrast, although some residual haze and slight loss of fine details remain.

C. Summary

Overall, the results demonstrate that the proposed implementation successfully captures the fundamental behavior of the GMAN architecture. While the performance is lower than the original model, the results are consistent with expectations given the constraints on dataset size, computational resources, and model complexity.

V. Conclusion

In this project, we implemented a simplified version of a model-agnostic convolutional neural network for single image dehazing. Unlike traditional methods that rely on explicit atmospheric models, the proposed approach directly learns the mapping from hazy to clean images using deep learning.

The results demonstrate the effectiveness of end-to-end learning for image restoration tasks. The model is capable of producing visually improved images and achieves reasonable performance in terms of standard evaluation metrics.

A. Limitations

- Limited dataset size due to computational constraints
- Simplified network architecture compared to original GMAN
- Limited training epochs

B. Future Work

- Training on larger datasets for improved performance
- Incorporating advanced architectures
- Extending the model to handle real-world haze conditions

References

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